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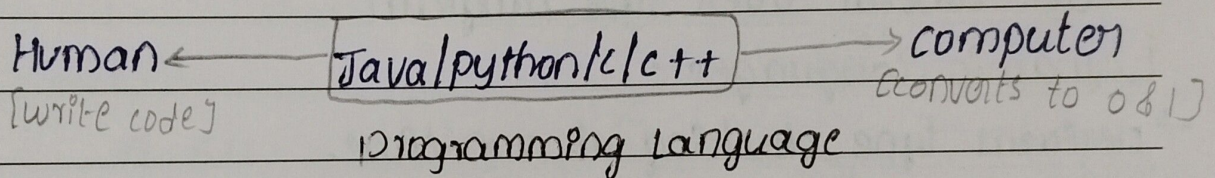
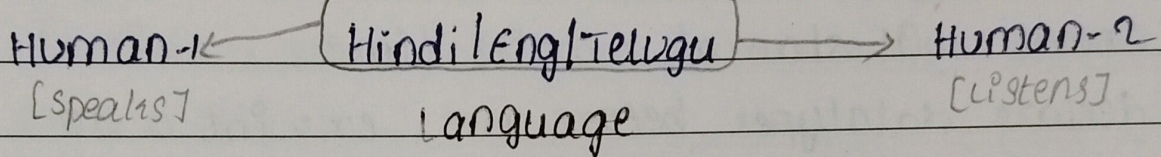
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Introduction to programming language

Computers understand only binary language i.e. 0's & 1's at very minute level.

What is programming language?

Is a formal language used to write instructions that a computer can understand & execute.



Types of PL:

1. procedural:- series of well-structured steps & procedures to compose a program.

→ contains a systematic order of statements functions & commands to complete a task.

2. Functional:- writing a program only in pure functions i.e, never modify variables but only create new ones as an output.

→ used in a situation where we have to perform lots of diff operations on the same set of data like ML.

3. object-oriented:

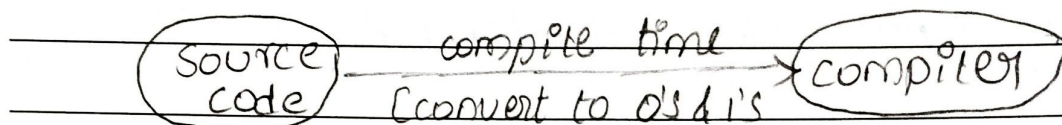
→ Revolves around the objects.

→ Code + data = objects (Data + methods)

→ developed to make it easier to develop, debug, reuse & maintain.

Static Language:

→ perform type checking at compile time.



→ errors will show at compile time

→ declare datatypes before use. ex: int a = 10

→ more control over the program

Dynamic languages:

→ perform type checking at run-time.



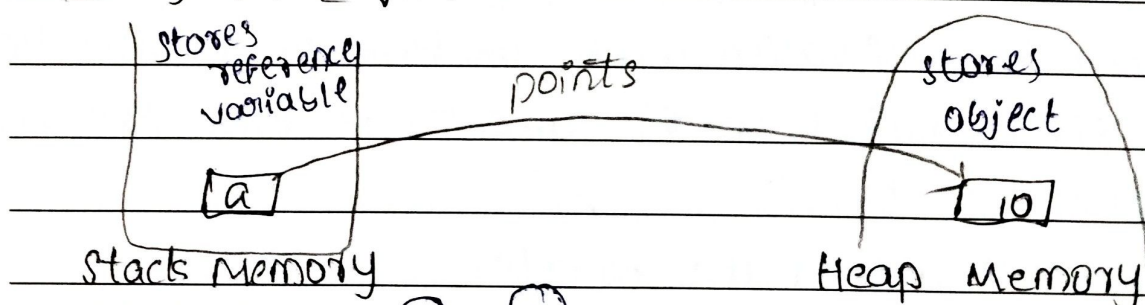
→ errors might not show till the programs run.

→ no need to declare datatype of variables.

Ex: a = 10; [language by itself figures out datatype]

→ saves time in writing code but might give error at runtime.

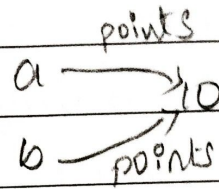
Memory Management



reference variable $\text{a} = \text{10}$ object

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Ex: $a = 10;$
 $b = a;$

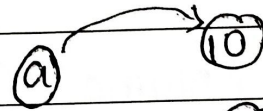


→ more than one reference variable can point towards one object.

→ If any of the reference variable changes the object then it is changed for all reference variables that points towards same object.

Ex: Initially,

$a = 10$



object with
no reference
variable

After,

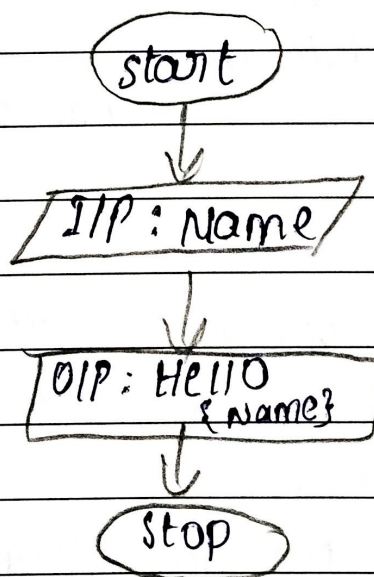
$a = 50$



Removed by
garbage
collector.

Flow of Program

Ex: Take a name & output Hello Name:



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Pseudo Code :

- It is just a way to write steps which is human readable format. [It is not a code].
- It is mainly meant for human reading not for machine reading.

Ex : Take above example to take a name & output

Step-1 : start Hello name :

Step-2 : Take input from user [name = input("enter name")]

Step-3 : Hello {name} [output]

Step-4 : stop.